

Matteo Luvra
Signal Processing
HW #4

1. Convolution & correlation
impulse;

$$\delta[n] \stackrel{\Delta}{=} \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$

a) $\delta = [1, 0, 0, 0]$ length $N=4$

signal: $x = [1, -1, 1, -1]$

$$x \otimes \delta = x$$

$$y[n] = (x \otimes h)[n] = \sum_{m=0}^3 x[m] h[(n-m) \bmod 4],$$

for $n = 0, 1, 2, 3$

$h = \delta$, so

$$y[n] = (x \otimes \delta)[n] = \sum_{m=0}^3 x[m] \delta[(n-m) \bmod 4]$$

$$\delta = [1, 0, 0, 0] \quad \text{-----}$$

$$\delta[0] = 1$$

$$\delta[1] = 0$$

$$\delta[2] = 0$$

$$\delta[3] = 0$$

$$y[0] = (x \otimes \delta)[0]$$

$$y[0] = \sum_{m=0}^3 x[m] \delta[(0-m) \bmod 4]$$

$$m = 0$$

$$(0-0) \bmod 4 = 0$$

$$\delta[0] = 1$$

$$x[0] \cdot \delta[0] = 1 \cdot 1 = 1$$

$$m = 2$$

$$(0-2) \bmod 4 = (-2) \bmod 4 = 2$$

$$\delta[2] = 0$$

$$x[2] \cdot \delta[2] = 1 \cdot 0 = 0$$

$$m = 1$$

$$-(0-1) \bmod 4 = (-1) \bmod 4 = 3$$

$$\delta[3] = 0$$

$$x[1] \cdot \delta[3] = (-1) \cdot 0 = 0$$

$$m = 3$$

$$(0-3) \bmod 4 = (-3) \bmod 4 = 1$$

$$\delta[1] = 0$$

$$x[3] \cdot \delta[1] = (-1) \cdot 0 = 0$$

$$y[0] = 1 + 0 + 0 + 0 = 1$$

$$y[1] = (x \otimes \delta)[1]$$

$$y[1] = \sum_{n=0}^3 x[n] \delta[(1-n) \bmod 4]$$

$$n=0$$

$$(1-0) \bmod 4 = 1$$

$$\delta[1] = 0$$

$$x[0] \cdot \delta[1] = 1 \cdot 0 = 0$$

$$n=1$$

$$(1-1) \bmod 4 = 0$$

$$\delta[0] = 1$$

$$x[1] \cdot \delta[0] = (-1) \cdot 1 = -1$$

$$n=2$$

$$(1-2) \bmod 4 = (-1) \bmod 4 = 3$$

$$\delta[3] = 0$$

$$x[2] \cdot \delta[3] = 1 \cdot 0 = 0$$

$$n=3$$

$$(1-3) \bmod 4 = (-2) \bmod 4 = 2$$

$$\delta[2] = 0$$

$$x[3] \cdot \delta[2] = (-1) \cdot 0 = 0$$

$$y[1] = 0 - 1 + 0 + 0 = -1$$

$$y[2] = \sum_{n=0}^3 x[n] \delta[(2-n) \bmod 4]$$

$$n=0$$

$$(2-0) \bmod 4 = 2$$

$$\delta[2] = 0$$

$$x[0] \cdot \delta[2] = 1 \cdot 0 = 0$$

$$n=1$$

$$(2-1) \bmod 4 = 1$$

$$\delta[1] = 0$$

$$x[1] \cdot \delta[1] = (-1) \cdot 0 = 0$$

$$m = 2$$

$$(2-2) \bmod 4 = 0$$

$$\delta[0] = 1$$

$$x[2] \cdot \delta[0] = 1 \cdot 1 = 1$$

$$m = 3$$

$$(2-3) \bmod 4 = (-1) \bmod 4 = 3$$

$$\delta[3] = 0$$

$$x[3] \cdot \delta[3] = (-1) \cdot 0 = 0$$

$$y[2] = 0 + 0 + 1 + 0 = 1$$

$$y[3] = (x \circledast \delta)[3]$$

$$y[3] = \sum_{m=0}^3 x[m] \delta[(3-m) \bmod 4]$$

$$m = 0$$

$$(3-0) \bmod 4 = 3$$

$$\delta[3] = 0$$

$$x[0] \cdot \delta[3] = 1 \cdot 0 = 0$$

$$m = 1$$

$$(3-1) \bmod 4 = 2$$

$$\delta[2] = 0$$

$$x[1] \cdot \delta[2] = (-1) \cdot 0 = 0$$

$$m = 2$$

$$(3-2) \bmod 4 = 1$$

$$\delta[1] = 0$$

$$x[2] \cdot \delta[1] = 1 \cdot 0 = 0$$

$$m = 3$$

$$(3-3) \bmod 4 = 0$$

$$\delta[0] = 1$$

$$x[3] \cdot \delta[0] = (-1) \cdot 1 = -1$$

$$y[3] = 0 + 0 + 0 - 1 = -1$$



$$y[0] = 1$$

$$y[1] = -1$$

$$y[2] = 1$$

$$y[3] = -1$$

$$\rightarrow x \otimes \delta = [1, -1, 1, -1] = x$$

$$16) \quad x \otimes \text{Shift}_1(\delta) = \text{Shift}_1(x), \text{ where} \\ \text{Shift}_{1,n}(x) \triangleq x[n-1]$$

$$y[n] = (x \otimes \text{Shift}_1(\delta))[n] = \sum_{m=0}^3 x[m] \text{Shift}_1(\delta)[n-m \bmod 4]$$

- computation of $\text{Shift}_1(\delta)$

$$\delta = [1, 0, 0, 0]$$

$$n=0 \rightarrow \text{Shift}_1(\delta)[0] = \delta[(0-1) \bmod 4] = \delta[-1 \bmod 4] = \delta[3] = 0$$

$$n=1 \rightarrow \text{Shift}_1(\delta)[1] = \delta[(1-1) \bmod 4] = \delta[0] = 1$$

$$n=2 \rightarrow \text{Shift}_1(\delta)[2] = \delta[(2-1) \bmod 4] = \delta[1] = 0$$

$$n=3 \rightarrow \text{Shift}_1(\delta)[3] = \delta[(3-1) \bmod 4] = \delta[2] = 0$$

as a vector:

$$\text{Shift}_1(\delta) = [0, 1, 0, 0]$$



$$y[0] = \sum_{m=0}^3 x[m] h[(0-m) \bmod 4]$$

$$m=0$$

$$h[(0-0) \bmod 4] = 0$$

$$h[0] = 0$$

$$x[0] \cdot h[0] = 1 \cdot 0 = 0$$

$$m=1$$

$$(0-1) \bmod 4 = (-1) \bmod 4 = 3$$

$$h[3] = 0$$

$$x[1] \cdot h[3] = (-1) \cdot 0 = 0$$

$$m=2$$

$$(0-2) \bmod 4 = (-2) \bmod 4 = 2$$

$$h[2] = 0$$

$$x[2] \cdot h[2] = 1 \cdot 0 = 0$$

$$m=3$$

$$(0-3) \bmod 4 = (-3) \bmod 4 = 1$$

$$h[1] = 1$$

$$x[3] \cdot h[1] = (-1) \cdot 1 = -1$$

$$y[0] = 0 + 0 + 0 - 1 = -1$$

$$y[1] = \sum_{m=0}^3 x[m] h[(1-m) \bmod 4]$$

$$m=0$$

$$(1-0) \bmod 4 = 1$$

$$h[1] = 1$$

$$x[0] \cdot h[1] = 1 \cdot 1 = 1$$

$$m=1$$

$$\text{index}(1-1) \bmod 4 = 0$$

$$h[0] = 0$$

$$x[1] \cdot h[0] = (-1) \cdot 0 = 0$$

$$m=2$$

$$(1-2) \bmod 4 = (-1) \bmod 4 = 3$$

$$h[3] = 0$$

$$x[2] \cdot h[3] = 1 \cdot 0 = 0$$

$$m=3$$

$$(1-3) \bmod 4 = (-2) \bmod 4 = 2$$

$$h[2] = 0$$

$$x[3] \cdot h[2] = (-1) \cdot 0 = 0$$

$$y[1] = 1 + 0 + 0 + 0 = 1$$

$$y[2] = \sum_{m=0}^3 x[m] h[(2-m) \bmod 4]$$

$$m=0$$

$$(2-0) \bmod 4 = 2$$

$$h[2] = 0$$

$$x[0] \cdot h[2] = 1 \cdot 0 = 0$$

$$m=1$$

$$(2-1) \bmod 4 = 1$$

$$h[1] = 1$$

$$x[1] \cdot h[1] = (-1) \cdot 1 = -1$$

$$m=2$$

$$(2-2) \bmod 4 = 0$$

$$h[0] = 0$$

$$x[2] \cdot h[0] = 1 \cdot 0 = 0$$

$$m=3$$

$$(2-3) \bmod 4 = (-1) \bmod 4 = 3$$

$$h[3] = 0$$

$$x[3] \cdot h[3] = (-1) \cdot 0 = 0$$

$$y[2] = 0 \cdot 1 + 0 + 0 = -1$$

$$y[3] = \sum_{n=0}^3 x[n] h[(3-n) \bmod 4]$$

$$n=0$$

$$(3-0) \bmod 4 = 3$$

$$h[3] = 0$$

$$x[0] \cdot h[3] = 1 \cdot 0 = 0$$

$$n=1$$

$$(3-1) \bmod 4 = 2$$

$$h[2] = 0$$

$$x[1] \cdot h[2] = (-1) \cdot 0 = 0$$

$$n=2$$

$$(3-2) \bmod 4 = 1$$

$$h[1] = 1$$

$$x[2] \cdot h[1] = 1 \cdot 1 = 1$$

$$n=3$$

$$(3-3) \bmod 4 = 0$$

$$h[0] = 0$$

$$x[3] \cdot h[0] = (-1) \cdot 0 = 0$$

$$y[3] = 0 + 0 + 1 + 0 = 1$$

$$\hookrightarrow y = x \otimes \text{shift}_1(\delta) = [-1, 1, -1, 1]$$

$$n=0$$

$$\hookrightarrow \text{shift}_1(x)[0] = x[(0-1) \bmod 4] = x[(-1) \bmod 4] = x[3] = -1$$

$$n=1$$

$$\hookrightarrow \text{shift}_1(x)[1] = x[(1-1) \bmod 4] = x[0] = 1$$

$$n=2$$

$$\hookrightarrow \text{shift}_1(x)[2] = x[(2-1) \bmod 4] = x[1] = -1$$

$$n=3$$

$$\hookrightarrow \text{shift}_1(x)[3] = x[(3-1) \bmod 4] = x[2] = 1$$

$$\text{shift}(x) = [-1, 1, -1, 1, 0]$$

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11) Find $(x \otimes [1, 1, 0, 0, \dots])_n$ using linearity of the convolution operator

$$h = [1, 1, 0, 0] = \delta + \text{shift}(\delta)$$

→ linearity $x \otimes (a + b) = x \otimes a + x \otimes b$

$$\hookrightarrow x \otimes h = x \otimes (\delta + \text{shift}(\delta)) = x \otimes \delta + x \otimes \text{shift}(\delta)$$

from 1(a) $\rightarrow x \otimes \delta = x$

1(b) $\rightarrow x \otimes \text{shift}(\delta) = \text{shift}(x)$

$$\hookrightarrow x \otimes h = x + \text{shift}(x)$$

$$x + \text{shift}(x)$$

$$x + \text{shift}(x) = [1+(-1), -1+1, 1+(-1), -1+1] = [0, 0, 0, 0]$$

$$\hookrightarrow (x \otimes [1, 1, 0, 0, \dots])_n = [0, 0, 0, 0]$$

1d)

$$x = [0, 0, 1, 0, 0, 0, 1, 0]$$

$$h = [1, 1, 1, 1, 1, 0, 0, 0]$$

$$y[n] = (x \otimes h)[n], \quad n = 0, \dots, 7$$

n	$(n-2)_{-8}$ ^{mod}	$h[(n-2)_{-8}]$	$(n-6)_{-8}$	$h[(n-6)_{-8}]$	$y[n]$
0	6	0	2	1	1
1	7	0	3	1	1
2	0	1	4	1	2
3	1	1	5	0	1
4	2	1	6	0	1
5	3	1	7	0	1
6	4	1	0	1	2
7	5	0	1	1	1

$$y = [1, 1, 2, 1, 1, 1, 2, 1]$$

$$(x \otimes h)[n] = [1, 1, 2, 1, 1, 1, 2, 1]$$

1e) $x = [1, 2, 3, 2]$ & $h = [3, -1, 2, 1]$

find $\text{conv}(x \otimes h)_n$ & correlation
 $(x * h)_n$

x is even

definition - of cyclic conv

$$y[n] = \sum_{m=0}^3 x[m] h[(n-m)_4]$$

$$y[0] = 1 \cdot 3 + 2 \cdot 1 + 3 \cdot 2 + 2 \cdot (-1) = 3 + 2 + 6 - 2 = 9$$

$$y[1] = 1 \cdot (-1) + 2 \cdot 3 + 3 \cdot 1 + 2 \cdot 2 = -1 + 6 + 3 + 4 = 12$$

$$y[2] = 1 \cdot 2 + 2 \cdot (-1) + 3 \cdot 3 + 2 \cdot 1 = 2 - 2 + 9 + 2 = 11$$

$$y[3] = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot (-1) + 2 \cdot 3 = 1 + 4 - 3 + 6 = 8$$

$$x \otimes h = [9, 12, 11, 8]$$

cyclic correlation $r = x \star h$

$$r[n] = \sum_{m=0}^3 x[m] h[(m+n)_4]$$

$$r[0] = 1 \cdot 3 + 2 \cdot (-1) + 3 \cdot 2 + 2 \cdot 1 = 3 - 2 + 6 + 2 = 9$$

$$r[1] = 1 \cdot (-1) + 2 \cdot 2 + 3 \cdot 1 + 2 \cdot 3 = -1 + 4 + 3 + 6 = 12$$

$$r[2] = 1 \cdot 2 + 2 \cdot 1 + 3 \cdot 3 + 2 \cdot (-1) = 2 + 2 + 9 - 2 = 11$$

$$r[3] = 1 \cdot 1 + 2 \cdot 3 + 3 \cdot (-1) + 2 \cdot 2 = 1 + 6 - 3 + 4 = 8$$

$$x \star h = [9, 12, 11, 8]$$

evenness

$$x = [1, 2, 3, 2], \quad x[1] = x[3], \quad x[2] = x[2],$$

$$\text{so } x[n] = x[-n \bmod 4] : x \text{ is even}$$

$$\hookrightarrow x \circledast h = x \star h = [9, 12, 11, 8]$$

2) Convolution Theorem

signal $x[n] = \cos(n\pi/2)$ & rect

window $w[n] = 1$ for $0 \leq n < 4$
and zero elsewhere.

period $0 \leq n < 8$

$$x[0] = \cos(0) = 1$$

$$x[1] = \cos(\pi/2) = 0$$

$$x[2] = \cos(\pi) = -1$$

$$x[3] = \cos(3\pi/2) = 0$$

$$x[4] = \cos(2\pi) = 1$$

$$x[5] = 0, \quad x[6] = -1, \quad x[7] = 0$$

So

$$x[n] = [1, 0, -1, 0, 1, 0, -1, 0]$$

and

$$w[n] = [1, 1, 1, 1, 0, 0, 0, 0]$$

$$X[k] = \sum_{n=0}^7 x[n] e^{-j2\pi kn/8}, \quad W[k] = \sum_{n=0}^7 w[n] e^{-j2\pi kn/8}, \quad k=0, \dots, 7$$

• DFT of $x[n] = X[k]$

only even n are non-zero

$$X[k] = x[0]e^0 + x[2]e^{-j2\pi k \cdot 2/8} + x[4]e^{-j2\pi k \cdot 4/8} + x[6]e^{-j2\pi k \cdot 6/8}$$

with $x[0]=1, x[2]=-1, x[4]=1, x[6]=-1$

$$X[k] = 1 - e^{-j\pi k/2} + e^{-j\pi k} - e^{-j3\pi k/2}$$

evaluating this for $k=0, \dots, 7$ gives:

$$X[2]=4, X[6]=4, \text{ and } X[k]=0 \text{ for } k \neq 2, 6$$

vec :

$$X[k] = [0, 0, 4, 0, 0, 0, 4, 0]$$

• DFT of $w[n]$: $W[k]$

since $w[n] = 1$ for $n = 0, 1, 2, 3$ and 0 otherwise:

$$W[k] = \sum_{n=0}^3 e^{-j2\pi kn/8} = 1 + e^{-j\pi k/4} + e^{-j\pi k/2} + e^{-j3\pi k/4}$$

$$W[0] = 4,$$

$$W[1] = (1-j)/(1+\sqrt{2}),$$

$$W[2] = 0,$$

$$W[3] = (1-j)(\sqrt{2}-1),$$

$$W[4] = 0,$$

$$W[5] = (1+j)(\sqrt{2}-1),$$

$$W[6] = 0,$$

$$W[7] = (1+j)(1+\sqrt{2})$$

$$W[k] = [4, (1-j)/(1+\sqrt{2}), 0, (1-j)(\sqrt{2}-1), 0, (1+j)(\sqrt{2}-1), 0, (1+j)(1+\sqrt{2})]$$

b) $x_w[n] = x[n] w[n]$

$$x_w[n] = x[n] w[n] = [1, 0, -1, 0, 0, 0, 0, 0]$$

$n=8$ DFT of $x_w[n]$

$$X_w[k] = \sum_{n=0}^7 x_w[n] e^{-j2\pi kn/8}$$

since only $n=0$ & $n=2$ are non-zero.

$$X_w[k] = 1 \cdot e^0 + (-1) \cdot e^{-j2\pi k2/8} = 1 - e^{-j\pi k/2}$$

$$e^{-j\pi/2} = -j, e^{-j\pi} = -1, e^{-j3\pi/2} = j, e^{-j2\pi} = 1$$

$$k=0 \rightarrow 1-1=0$$

$$k=1 \rightarrow 1-(-j) = 1+j$$

$$k=2 \rightarrow 1-(-1) = 2$$

$$k=3 \rightarrow 1-(j) = 1-j$$

$$k=4 \rightarrow 1-1=0$$

$$k=5 \rightarrow 1+j$$

$$k=6 \rightarrow 2$$

$$k=7 \rightarrow 1-j$$

$$X_w[k] = [0, 1+j, 2, 1-j, 0, 1+j, 2, 1-j]$$

Σ ()

$$\begin{aligned}
 (X * W)[k] &= \sum_{n=0}^7 X[n] W[(k-n)_8] = \\
 &= X[2] W[(k-2)_8] + X[6] W[(k-6)_8] = \\
 &= 4 W[(k-2)_8] + 4 W[(k-6)_8]
 \end{aligned}$$

so

$$\begin{aligned}
 \frac{1}{8} (X * W)[k] &= \frac{1}{8} (4 W[(k-2)_8] + 4 W[(k-6)_8]) = \\
 &= \frac{1}{2} (W[(k-2)_8] + W[(k-6)_8])
 \end{aligned}$$

$W[k]$ values from $Z(a)$ are equal for
 $k = 0 \dots 7 = Z(b)$

$$\frac{1}{8} (X * W)[k] = X_w[k] \quad \forall k \quad \swarrow \text{all } k$$

$$\hookrightarrow X_w[k] = \frac{1}{8} (X * W)[k],$$

$\hookrightarrow$ DFT of the product $x[n]w[n]$
 equals $\frac{1}{N}$ times the circular
 convolution of $X[k]$ and $w[k]$

2d) $N=8$ IDFT of conv. of $X[k]$ & $w[k]$

let

$X[k] = \text{DFT of } x[n]$

$w[k] = \text{DFT of } w[n]$

$(X * w)[k] = \text{circular convolution in } k \text{ of } X \text{ \& } w$

$N=8$ IDFT is:

$$z[n] = \frac{1}{8} \sum_{k=0}^7 (X * w)[k] e^{j2\pi kn/8}$$

↳ $x_w[n]$ is the IDFT of $X_w[k]$ with usual $1/8$ factor.

$$2b \rightarrow x_w[n] = [1, 0, -1, 0, 0, 0, 0, 0]$$

$$\hookrightarrow z[n] = \frac{b}{8} x_w[n] = [0, 0, -0, 0, 0, 0, 0, 0]$$

$$\hookrightarrow [0, 0, -0, 0, 0, 0, 0, 0]$$

3) LTI, FIR/IIR Filters:

$$3a) y[n] = x[n] + 2x[n-1] + x[n-2]$$

• linearity

$$T\{x[n]\} = x[n] + 2x[n-1] + x[n-2]$$

input $a x[n]$, then

$$T\{a x[n]\} = a x[n] + 2 a x[n-1] + a x[n-2] = \\ a T\{x[n]\}$$

input $x_1[n] + x_2[n]$, then

$$T\{x_1 + x_2\} = (x_1 + x_2)[n] + 2(x_1 + x_2)[n-1] + \\ (x_1 + x_2)[n-2] = T\{x_1\} + T\{x_2\}$$

linear

- Invariance

$$x_1[n] = x[n - n_0] \Rightarrow y[n - n_0]$$

↳ time invariant

- FIR vs IIR

only uses $x[n]$, $x[n-1]$, $x[n-2]$. No $y[\cdot]$

↳ FIR

36)

$$\text{system: } y[n] = x[n] \times x[n-1]$$

$$\text{linearity: } T(ax_1 + bx_2) \neq aT(x_1) + bT(x_2)$$

↳ non-linear

time-invariance:

$$T\{x[n - n_0]\} = x[n - n_0] \times [n - 1 - n_0] = y[n - n_0]$$

↳ time invariant

FIR/IIR

↳ depends only on finite input samples,
not feedback

↳ FIR

3c)

system:

$$y[n] = x[n] + n$$

linearity:

$$T(x_1 + x_2) = x_1[n] + x_2[n] + n \neq T(x_1) + T(x_2)$$

↳ non-linear

Time invariance check:

shift by n_0 :

$$T(x_1 + x_2) = x_1[n] + x_2[n] + n$$

$\neq$

$$T(x_1) + T(x_2)$$

vs.

$$y[n - n_0] = x[n - n_0] + (n - n_0)$$

not equal $\rightarrow$ Time varying

FIR/IIR

no past outputs, finite input dependence

FIR

3d)

system: $y[n] = x[n] - y[n-1]$

linearity:

equation is linear in x & y

linear

time invariance:

shift by n_0 :

$$x[n-n_0] - y[n-1-n_0] = y[n-n_0]$$

↳ time invariant

FIR/IIR:

depends on past outputs $y[n-1]$

→ IIR

4) Difference Equations

$$y[n] = x[n] + 0.9y[n-1]$$